Lovro Rabuzin

lovro-rabuzin | LRabuzin | Zürich, Switzerland

Education

Jul 2024 – Ongoing
***ETH Zürich, Institute of Machine Learning* - PhD Candidate in Computer Science**
Topic: Novel algorithms on spatial omics for cancer biomarker discovery and hypothesis generation
PI: *Valentina Boeva*

Sep 2021 – May 2024
***ETH Zürich* - MSc in Computer Science** GPA: 5.66/6.0
Major: *Machine Intelligence* **Minor:** *Data Management*
Master thesis: Graph Neural Network-Driven Analysis of Cellular Organization for Non-Small Cell Lung Cancer Precision Oncology

Jul 2018 – Jul 2021
***FER, University of Zagreb, Croatia* - BSc in Computing** GPA: 4.88/5.0
Bachelor thesis: Semantic Segmentation of Neoplasms in Magnetic Resonance Images using Deep Convolutional Neural Networks

Work Experience

Sep 2022 – Jan 2023
***Unit8 SA* - Data Engineer Intern**
- Built a library for analysis of **Spark** plans on column level for tracking data lineage using **Python**
- Maintained and upgraded a data ingestion, processing and analysis pipeline for measuring business key performance indicators using **Azure**, **Databricks** and **PowerBI**

Teaching Experience

Feb 2026 – Aug 2026
***ETH Zürich* - Teaching Assistant - Big Data for Engineers**
- Held tutorials and exercise sessions for students taking the course
- Wrote and graded parts of the exam

Sep 2024 – Feb 2026
***ETH Zürich* - Head Teaching Assistant - Machine Learning for Genomics**
- Taught a lecture on methods for analysis of spatial omics data
- Held tutorials and exercise sessions for students taking the course
- Administrated and graded student paper presentations
- Wrote and graded parts of the exam

Feb 2025 – Aug 2025
***ETH Zürich* - Teaching Assistant - Computational Intelligence Lab**
- Held tutorials and exercise sessions for students taking the course
- Graded student research papers
- Wrote and graded parts of the exam

Conference Presentations

Dec 2025
***Exploring Augmentation-Driven Invariances for Graph Self-supervised Learning in Spatial Omics* - Workshop on Unifying Representations in Neural Models @ NeurIPS**
- Poster presentation showcasing results of a paper accepted to PMLR of the UniReps workshop.

Sep 2025
***GRASS-MIL: Graph-based Representation and Analysis of Spatial Structures with Multiple Instance Learning* - Basel Computational Biology Conference**
- Poster presentation showcasing preliminary results of ongoing PhD research

Sep 2024
***Spatial Omics Analysis for Hypothesis Generation Using Graph Neural Networks to Explore Cellular Organization in Non-Small Cell Lung Cancer* - scverse Conference, Munich, Germany**
- Poster presentation showcasing preliminary results of ongoing PhD research

Jun 2024
***Spatial Omics Analysis for Hypothesis Generation Using Graph Neural Networks to Explore Cellular Organization in Non-Small Cell Lung Cancer* - Spatial Omics Conference, Ghent, Belgium**
- Poster presentation showcasing preliminary results of ongoing PhD research

Conference Service

Sep 2025
***Reviewer* - Workshop on Unifying Representations in Neural Models @ NeurIPS**
- Reviewed submissions for the UniReps Workshop at NeurIPS 2025

Publications

Dec 2025 | ***Exploring Augmentation-Driven Invariances for Graph Self-supervised Learning in Spatial Omics* - Workshop on Unifying Representations in Neural Models @ NeurIPS**
- **Lovro Rabuzin**, Michel Tarnow, Valentina Boeva
- `https://openreview.net/forum?id=3riPC1Y2Jb`

Dec 2025 | ***DeepCAST-GWAS: Improving the Discovery of Genetic Associations Using Deep Learning-Based Regulatory SNP Prioritization* - bioRxiv pre-print**
- **Lovro Rabuzin**, Konstantin Heep, Sophie Sigfstead, Valentina Boeva
- `https://www.biorxiv.org/content/10.1101/2025.11.27.690924v1`

Sep 2024 | ***NeRFmentation: NeRF-based Augmentation for Monocular Depth Estimation* - Workshop on Synthetic Data for Computer Vision @ NeurIPS**
- Casimir Feldmann, Niall Siegenheim, Nikolas Hars, **Lovro Rabuzin**, Mert Ertugrul, Luca Wolfart, Marc Pollefeys, Zuria Bauer, Martin R. Oswald
- `https://doi.org/10.48550/arXiv.2401.03771`

Scholarships, Honours and Grants

Dec 2024 – Dec 2027 | ***ETH Grant* - ETH Zürich**
- Supporting PhD research on *Decoding NSCLC: novel algorithms for spatial omics to reveal key structural patterns for cancer biomarker discovery and hypothesis generation*

Oct 2024 | ***ETH Medal* - ETH Zürich**
- Awarded in recognition of an outstanding Master's thesis at ETH Zürich
- Granted to top Master's theses demonstrating exceptional academic merit and innovation

Oct 2018 – Jul 2021 | ***STEM Scholarship* - Croatian Ministry of Science and Education**
- Scholarship awarded to students with outstanding grades in university entrance exams
- In subsequent years, awarded to top-performing students in their respective undergraduate programmes

Nov 2019 – Nov 2020 | ***Dean's list award "Josip Lončar"* - FER, University of Zagreb**
- For excellent achievements in the second year of undergraduate studies
- Awarded to the top 1% of all students in the study programme

Languages

Croatian	Native speaker	
English	CEFR level C2 - TOEFL score 119	22 Nov 2020
German	CEFR level C1 - DSD II	14 Feb 2018

Skills

Programming and Markup Languages

Python, C/C++, R, LaTeX, Java, Go

Machine Learning Frameworks and Tools

PyTorch, Weights & Biases, NumPy, Scikit-Learn, Pandas

Data Management Technologies

SQL, Spark, Palantir Foundry, Microsoft Azure, AWS, Neo4j

User Software

Git, Unix shell